BBO capstone project presentation template

Develop a concise presentation that captures:

1. **An overview of your BBO approach**
   - Describe what you are trying to achieve or discover through the BBO capstone project. Focus on the core objective of your optimisation process, using clear and accessible language.
   - Briefly describe your overall strategy and how your process fits together.

Core Objective
- Maximize total output across eight unknown functions with only 13 attempts per function
- Each query is expensive - cannot waste opportunities learning what works
- Challenge is discovering patterns with minimal data while functions remain black boxes
-One function dominates 99% of total score, requiring strategic resource allocation
-Ultimate goal: achieve highest possible score by final submission through intelligent sequential
decision-making

Overall Strategy
-Hybrid approach combining machine learning models with human pattern recognition
-Gaussian Processes predict promising regions based on smoothness assumptions
-Random Forest identifies which variables matter most for each function
-Bayesian Optimization balances finding new peaks versus refining known ones
-When models fail at boundaries, switch to empirical pattern analysis - looking for obvious gaps in
tested values
-Process evolves weekly: analyze data, train models, compare predictions against patterns,
decide, submit, learn
-Early weeks focused on exploration mapping the landscape
-Middle weeks refined promising regions discovered
-Late weeks exploited proven discoveries while testing specific hypotheses
-Allocate 75% effort to high-value function, accept losses on noisy or plateaued functions

2. **How your strategy has evolved**
   - Describe the key changes in your approach since your early rounds.
   - What influenced these changes – data trends, feedback, model performance or intuition?
   - Identify which principles or heuristics now guide your query decisions.

Key Changes Since Early Rounds
-Weeks 1-4 used pure exploration with random sampling and grid search
-Week 5 breakthrough when Random Forest revealed pushing two variables to 1.0 triggered 40x
output jump
-This regime shift was invisible to Gaussian Processes because they assume smoothness
-Week 8 disaster taught hardest lesson - changed critical variable, lost 1,742 points instantly
-Now only change one variable at a time for clean causal attribution
-Learned short length scales mean 'critical boundary, don't touch'
not 'optimize this'
-Shifted from trusting models everywhere to hybrid model-empirical approach
-Models work for smooth regions, empirical patterns work for boundaries

What Influenced Changes
-Data trends showed plateaus around certain values requiring boundary testing
-Model performance degraded at edges - predicted improvements that empirically declined
-Week 10 accidental duplicate revealed which functions are deterministic versus stochastic
-Enabled proper noise handling - added noise kernels to some, abandoned modeling on others
-Week 11 breakthrough came from noticing empirical gap tested many boundary values except
1.0
-Gained 368 points from obvious test that models couldn't suggest
-Week 8 failure from trusting model signal without understanding what it meant

Guiding Principles Now
-Never change variables with ARD length scales below 0.1, these are critical boundaries
-Test empirical gaps before trusting model extrapolation at function edges
-Single-variable isolation prevents multi-variable disasters
-Accept that highly stochastic functions cannot be modeled reliably
-Focus 75% effort on high-value functions still improving
-Accept losses on plateaued or noisy functions rather than wasting queries
-Trust data patterns over model predictions when they conflict at boundaries
-Use models for smooth interpolation, use patterns for regime shifts and peak

3. **Patterns, data and insights**
   - Highlight the most meaningful trends you've observed across your data points or model evaluations.
   - Identify which variables or behaviours seem to influence results the most.
   - Explain how these observations have shaped your understanding of the search process.

Most Meaningful Trends
-F5 has two distinct regimes low around 137 points and high around 6,500+ points
-Transition happens when x1 and x3 both reach approximately 1.0
-Within high regime, x2 optimum shifted from 0.853 to 0.920 when x4 reached 1.0. This reveals boundary conditions interact - regime-dependent optimization
-Week 10 duplicates characterized noise: five functions deterministic, one highly stochastic, two
moderate
-F7 shows clear peak at x6 equals 0.734 - values above or below decline symmetrically
-F4 is symmetric quadratic bowl centered at all values 0.5

Most Influential Variables
-F5 x1 and x3 are non-negotiable boundaries that must equal 1.0
-F5 x2 is primary tunable parameter with non-monotonic behavior
-F5 x4 unlocks regime shifts at 1.0 boundary
-F7 x6 dominates all other dimensions in importance
-F4 all variables equally important due to symmetry
-F2 shows 80% variance making all variables essentially random
-Boundary adherence emerges as universal pattern across functions

How Observations Shaped Understanding
-Learned models fail at discontinuities and regime transitions
-Smoothness assumption makes Gaussian Processes excellent interpolators but poor at detecting
jumps
-Empirical pattern recognition outperformed sophisticated optimization at boundaries
-Best gains came from hybrid thinking - models for smooth regions, patterns for edges
-High-dimensional functions severely undersampled - F8 explored only 0.00000001% of space
-Clustering at boundaries often represents true structure, not search bias
-Single-variable isolation enables understanding which dimensions actually matter

4. **Decision-making and iteration**
   - Explain how you balance exploration and exploitation.
   - Describe one or two examples of strategic decisions. What worked, what didn't, and why?
   - Reflect briefly on how you handle uncertainty or adapt when results don't match expectations.

Exploration-Exploitation Balance
-Early weeks 1-4: 80% exploration mapping unknown landscape
-Weeks 5-7: 50-50 after finding high regime
-Weeks 8-11: 70% exploitation with careful boundary exploration after disaster
-Week 12: 40% exploration testing regime-shift hypothesis
-Week 13: 95% exploitation locking in all proven bests-
-Balance shifted as data accumulated and uncertainty decreased
-Cannot afford pure exploration late with only 13 total queries

Strategic Decision Examples
-Success Week 11: Noticed empirical gap in tested boundary values, x4 never tested at 1.0
-GP couldn't suggest this because it prefers smooth interpolation
-Testing obvious boundary gained 368 points - largest single improvement
-Success Week 12: Hypothesized x4 equals 1.0 enables higher x2 optimum
-Tested x2 equals 0.920 with x4 equals 1.0, gained 815 points
-Failure Week 8: Changed x1 from 1.0 to 0.855 because it had shortest length scale
-Lost 1,742 points - misunderstood signal means high sensitivity not optimization opportunity

Handling Uncertainty
-When results don't match expectations, first check if assumption was wrong
-Use single-variable isolation to identify what caused the change
-If function shows high noise through duplicates, abandon modeling and use best historical value
-When GP predictions diverge from empirical patterns, trust the data
-Models assume smoothness that may not exist at regime boundaries
-Build in safety nets - Week 12 aggressive test protected by Week 11 locked floor
-Accept that some functions cannot be optimized reliably given constraints

5. **Next steps and reflection**
   - Outline your planned actions or experiments to improve the model's performance.
   - Include two or three sentences connecting your work to the broader ML landscape.
   - Add a short statement on how you'd communicate your results to a non-technical audience or stakeholder.

Planned Actions
-If continuing, would test F5 interior space - x2 and x4 values 0.3-0.8 completely unexplored in
high regime
-Run deliberate duplicate tests on moderate-noise functions to better quantify uncertainty
-Test if F8 has regime shifts like F5 - only explored 0.00000001% of 8D space
-Investigate whether F3 near-constraint boundary has unexplored potential
-Would implement multi-fidelity optimization using cheap approximations to guide expensive queries
-Explore whether ensemble acquisition functions combining Expected Improvement and UCB improve boundary detection

Connection to Broader ML
-Project demonstrates core challenges in real-world machine learning - optimizing expensive black
boxes with limited budgets like hyperparameter tuning, A/B testing, simulation-based design
-Regime shift discovery mirrors concept drift in production systems where models trained on one
distribution fail when underlying process changes
-Hybrid model-empirical approach reflects growing recognition that purely algorithmic solutions
miss patterns obvious to human judgment, suggesting value in human-in-the-loop optimization

Communicating to Non-Technical Audience
-Imagine tuning a complex machine with eight interconnected dials but you cannot see inside it
you only get a score after each adjustment
-Started by trying random settings to learn the landscape, then discovered pushing two specific
dials all the way up triggered massive performance jump - like finding hidden turbo mode
-Achieved 50x improvement by combining computer predictions with pattern recognition about
what hasn't been tried yet
-Computer models work great for smooth relationships but miss sudden jumps - that's where human judgment made the difference
-Final result: 4,900% improvement from baseline